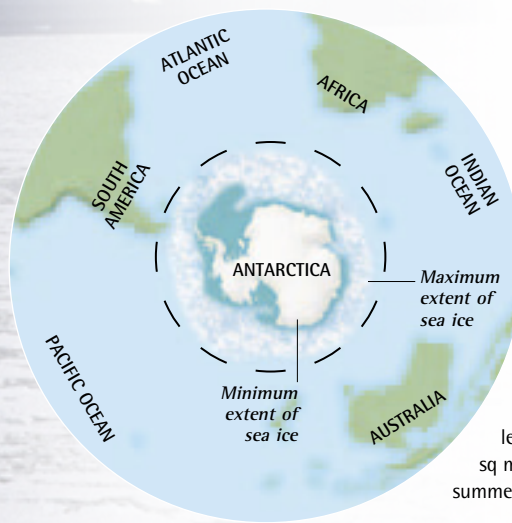




◀ EXPANDING ICE

In winter, the area of sea ice around Antarctica increases from about 4 million to 1.5 to 8.5 million sq miles (22 million sq km). In the north, sea ice covers most of the Arctic Ocean in winter, but it shrinks in summer to leave as little as 2.3 million sq miles (6 million sq km) of summer ice near the North Pole.

THE WATER BELOW



Sea ice insulates the water below from extremely low winter air temperatures, but there is virtually no light because of the ice cover and the dark polar winter. Salt expelled by ice formation makes the water more dense, so it sinks. This forces deeper, nutrient-rich water to the surface, fueling the growth of marine life when the ice thaws. The sinking water also drives deep-water currents, and draws surface currents toward the polar regions.



▲ ICEBERGS

Where glaciers reach the sea, big slabs of ice break off and float away as icebergs. In 2000, an iceberg the size of the state of Connecticut broke away from the Ross Ice Shelf in Antarctica. Most icebergs are smaller, but since 90 percent of their bulk is underwater, they are all bigger than they look.